

Rendering Comparison: DMLex Version 1.0 OASIS Standard, DocBook and Markdown editions

24 September 2026. OASIS TC Administration, for the LEXIDMA TC (TCADMIN-4742).

Summary

This report compares the Markdown edition of DMLex Version 1.0 OASIS Standard, rendered through the OASIS publication pipeline, with the edition OASIS published from the TC's DocBook source. Each figure puts the published edition on the left and the Markdown edition on the right, at the same place in the document.

Measure	Published (DocBook)	Markdown edition
Words compared (tokens)	61,253	61,225, 0 unexplained differences
Numbered headings	311	311
Code blocks	316	316, all identical
Internal links	3,166	3,166, none broken
Figures	49 at 566.9px	49 at 567px
PDF pages	218	194
PDF body text	10pt	10pt
PDF code	10pt	9pt
PDF footer	8pt	8pt

The text is the same. The Markdown edition takes the look of the OASIS Markdown template. The PDF runs to more pages because code blocks carry a border and padding, and notes render as shaded blocks.

How the comparison was made

- **Text.** [tools/docbook-to-markdown/verify_md.py](#) reduces both editions to visible text and aligns them word by word against the live page at docs.oasis-open.org. It also compares code blocks byte for byte, headings in order, every internal anchor and every image. Eleven accepted deviations are listed in [allow.json](#), each with its reason.
- **Rendering.** [tools/publication-assurance/render.sh](#) runs the OASIS pipeline: pandoc with the OASIS stylesheet and the OASIS post-processor for HTML, then the OASIS PDF preprocessor and headless Chrome for PDF. It stages the result at its docs.oasis-open.org path and runs oasis-pub-check v1.4.2.
- **Pictures.** [tools/publication-assurance/compare.mjs](#) scrolls both HTML editions to the same anchor and captures the viewport. PDF pages were matched by their text and rasterised at 108 dpi. Type sizes are read from the PDF text layer, as the size of each run of text weighted by its characters.

Defects found and fixed

Three rendering defects were found by these comparisons and fixed.

- **Figure width.** Every figure in the DocBook source is 16cm wide, and the converter dropped the width, so the Appendix D UML diagram ran off the page. The converter now carries the width through.

- **PDF scale and footer.** One long inline code path in section 3.2.1 could not wrap, and Chrome shrank every page of the PDF to fit it: 12pt text printed at about 7.5pt. The OASIS PDF preprocessor now lets inline code wrap (publication-assurance v1.4.2). The PDF also now carries the published footer on every page.
- **PDF type size** (fixed 25 September). The OASIS Markdown stylesheet sets 12pt body text and no print size, and headless Chrome prints it at 12pt, against 10pt in the published PDF. The first version of this report compared word-box heights, which include line spacing, and scored the two as equal. The OASIS PDF preprocessor now sets the print sizes in points (publication-assurance v1.5.0): body 10pt, code 9pt, tables 9pt, footer 8pt. The PDF is 194 pages, down from 228.

Differences that remain

These come from the OASIS Markdown template, not from the content.

- The cover labels the URIs "This stage", "Previous stage" and "Latest stage", where the DocBook stylesheet says "version".
- Notes render as shaded blocks with a rule, where DocBook uses a "Note" heading.
- The table of contents is a nested list, and object-type names in it are in body type rather than monospace.
- Code blocks have a grey background with a border, where DocBook uses lilac with no border.
- The Appendix D UML diagram is regenerated from the TC's own source by Graphviz, so boxes sit in different places. The classes, attributes and relations are the same.

Published HTML (DocBook)	Markdown edition HTML (OASIS pipeline)
 Data Model for Lexicography (DMLex) Version 1.0 OASIS Standard 29 April 2025 Specification URIs This version: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/dmlex-v1.0-os.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/dmlex-v1.0-os.pdf (Authoritative) Previous version: https://docs.oasis-open.org/lexidma/dmlex/v1.0/cs01/dmlex-v1.0-cs01.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/cs01/dmlex-v1.0-cs01.pdf (Authoritative) Latest version: https://docs.oasis-open.org/lexidma/dmlex/v1.0/dmlex-v1.0.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/dmlex-v1.0.pdf (Authoritative) Technical Committee: OASIS Lexicographic Infrastructure Data Model and API (LEXIDMA) TC Chair: Michal Měchura (michmech@mail.muni.cz), Masaryk University Editors: David Filip (glorfindel@mail.muni.cz), Masaryk University Miloš Jakubiček (milos.jakubicek@sketchengine.eu), Lexical Computing Vojtěch Kovář (vojcek@mail.muni.cz), Masaryk University Simon Krek (simon.krek@ijs.si), Jozef Stefan Institute John McCrae (john.mccrae@universityofgalway.ie), University of Galway Michal Měchura (michmech@mail.muni.cz), Masaryk University Additional artifacts: This prose specification is one component of a Work Product that also includes declarative validation artifacts accessible from https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/: - XML: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/XML/ - JSON: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/JSON/ - RDF: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/RDF/ Informative copies of third party schemas are provided: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/informativeCopiesOf3rdPartySchemas/ Declared namespaces: This specification declares one or more namespaces. Namespace isn't considered an XML specific feature in this serialization independent specification. The core and modules namespace http://docs.oasis-open.org/lexidma/ns/dmlex-1.0 Key words: The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY, and OPTIONAL in this document are to be interpreted as described in BCP 14 [RFC2119] and [RFC8174] if, and only if, they appear in all capitals, as shown here. Abstract: This document defines the 1st version of a data model in support of the high-priority technical goals described in the LEXIDMA TC's charter, including: - A serialization-independent Data Model for Lexicography (DMLex) - An XML serialization of DMLex - A JSON serialization of DMLex - A relational database serialization of DMLex - An RDF serialization of DMLex - An informative NVH serialization of DMLex Status:	 Data Model for Lexicography (DMLex) Version 1.0 OASIS Standard 29 April 2025 This stage: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/dmlex-v1.0-os.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/dmlex-v1.0-os.pdf (Authoritative) Previous stage: https://docs.oasis-open.org/lexidma/dmlex/v1.0/cs01/dmlex-v1.0-cs01.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/cs01/dmlex-v1.0-cs01.pdf (Authoritative) Latest stage: https://docs.oasis-open.org/lexidma/dmlex/v1.0/dmlex-v1.0.html https://docs.oasis-open.org/lexidma/dmlex/v1.0/dmlex-v1.0.pdf (Authoritative) Technical Committee: OASIS Lexicographic Infrastructure Data Model and API (LEXIDMA) TC Chair: Michal Měchura (michmech@mail.muni.cz), Masaryk University Editors: David Filip (glorfindel@mail.muni.cz), Masaryk University Miloš Jakubiček (milos.jakubicek@sketchengine.eu), Lexical Computing Vojtěch Kovář (vojcek@mail.muni.cz), Masaryk University Simon Krek (simon.krek@ijs.si), Jozef Stefan Institute John McCrae (john.mccrae@universityofgalway.ie), University of Galway Michal Měchura (michmech@mail.muni.cz), Masaryk University Additional artifacts: This prose specification is one component of a Work Product that also includes declarative validation artifacts accessible from https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/: - XML: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/XML/ - JSON: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/JSON/ - RDF: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/RDF/ Informative copies of third party schemas are provided: https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/informativeCopiesOf3rdPartySchemas/ Declared namespaces: This specification declares one or more namespaces. Namespace isn't considered an XML specific feature in this serialization independent specification. The core and modules namespace http://docs.oasis-open.org/lexidma/ns/dmlex-1.0 Abstract: This document defines the 1st version of a data model in support of the high-priority technical goals described in the LEXIDMA TC's charter, including: - A serialization-independent Data Model for Lexicography (DMLex) - An XML serialization of DMLex - A JSON serialization of DMLex

Figure 1: Cover and front matter

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Figure 2: Table of Contents

Published HTML (DocBook)	Markdown edition HTML (OASIS pipeline)
3.4 entry Represents a dictionary entry. An entry contains information about one headword. <i>Property of</i> lexicographicResource <i>Properties</i> - headword REQUIRED (exactly one) and UNIQUE (in combination with other unique properties if present). Normalised string. The entry's headword. partOfSpeech OPTIONAL (zero or more) and UNIQUE (in combination with other unique properties if present). - homographNumber OPTIONAL (zero or one) and UNIQUE (in combination with other unique properties if present). Number. The entry's homograph number, as a guide to distinguish entries with the same headword. label OPTIONAL (zero or more). pronunciation OPTIONAL (zero or more). inflectedForm OPTIONAL (zero or more). sense OPTIONAL (zero or more). Note DMLex Core does not have a concept of "subentry". To model subentries (i.e. entries inside entries) in a lexicographic resource, object types from the Linking Module should be used. Note The headword can be a single word, a multi-word expression, or any expression in the source language which is being described by the entry. Note DMLex allows only one headword per entry. Things such as variant headwords do not exist in DMLex. However, the DMLex Linking Module does make it possible to represent the existence of variants by treating them as separate headwords of separate entries, and linking the entries using a type of link which will cause the entries to be placed together when shown to human users. See Section A.1.16 "Modelling variants", for an example using the English words "colour" and "color". Note DMLex allows entries to contain zero senses. An entry with zero senses is an entry which declares the existence of its headword and, optionally, gives some information about the headword's morphosyntactical properties (part of speech, inflected forms etc.) but does not say anything about its semantics, about its meaning. Such entries can exist as entries which are work in progress, or as entries whose only purpose is to serve as members in entry-to-entry relations (defined in the DMLex Linking Module) such as spelling variants (example: Section A.1.16 "Modelling variants"). Note Entries in DMLex do not have an explicit listing order. An application can imply a listing order from a combination of the headword and the homograph number, such that the headword is the primary sorting key and the homograph number (for entries that have one) is the secondary sorting key. Note Ideally, each entry should have exactly one part-of-speech label. However, DMLex allows more than one partOfSpeech in entry in order to allow for exceptional cases when the lexicographer has decided to treat multiple part-of-speech readings of a headword in a single entry. Example: English words which denote nationalities ("Czech", "German") and which can function both as nouns and as adjectives.	3.4 entry Represents a dictionary entry. An entry contains information about one headword. Property of lexicographicResource Properties - headword REQUIRED (exactly one) and UNIQUE (in combination with other unique properties if present). Normalised string. The entry's headword. partOfSpeech OPTIONAL (zero or more) and UNIQUE (in combination with other unique properties if present). - homographNumber OPTIONAL (zero or one) and UNIQUE (in combination with other unique properties if present). Number. The entry's homograph number, as a guide to distinguish entries with the same headword. label OPTIONAL (zero or more). pronunciation OPTIONAL (zero or more). inflectedForm OPTIONAL (zero or more). sense OPTIONAL (zero or more). Note DMLex Core does not have a concept of "subentry". To model subentries (i.e. entries inside entries) in a lexicographic resource, object types from the Linking Module should be used. 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<i>Example 6. XML</i>	<i>Example 6. XML</i>
<pre><entry homographNumber="..."> <headword>...</headword> <partOfSpeech.../> <label.../> <pronunciation.../> <inflectedForm.../> <sense.../> </entry></pre>	<pre><entry homographNumber="..."> <headword>...</headword> <partOfSpeech.../> <label.../> <pronunciation.../> <inflectedForm.../> <sense.../> </entry></pre>
<i>Example 7. JSON</i>	<i>Example 7. JSON</i>
<pre>{ "headword": "...", "homographNumber": "...", "partOfSpeech": [...], "labels": [...], "pronunciations": [...], "inflectedForms": [...], "senses": [...] }</pre>	<pre>{ "headword": "...", "homographNumber": "...", "partOfSpeech": [...], "labels": [...], "pronunciations": [...], "inflectedForms": [...], "senses": [...] }</pre>
<i>Example 8. RDF</i>	<i>Example 8. RDF</i>

Figure 3: Section 3.4 entry: properties and notes

A.1.16 Modelling variants

We have two entries in our lexicographic resource, one for the headword "colour" and one for the headword "color". We want to create a relation to represent the fact that these are spelling variants. One of the entries is fully fleshed-out (has a sense with a definition, an example etc) while the other one is only skeletal: its only purpose is to serve as the origin of a navigable link to the other entry.

Example A.79. Pseudocode

```
- lexicographicResource (uri: http://example.com, langCode: en)
- entry (id: colour, headword: colour)
  - partOfSpeech (tag: n)
  - label (tag: europeanSpelling)
  - sense (id: colour-1)
    - definition (text: red, blue, yellow etc.)
    - example (text: What is your favourite colour?)
- entry (id: color, headword: color)
  - partOfSpeech (tag: n)
  - label (tag: americanSpelling)
- relation (type: variants)
  - member (ref: colour)
  - member (ref: color)
- relationType (type: variants, description: words which differ
  only in spelling)
- memberType (type: entry, min: 2, hint: navigate)
```

Example A.80. XML

```
<lexicographicResource uri="http://example.com" langCode="en"
xmlns="http://docs.oasis-open.org/lexidma/ns/dmlex-1.0">
  <entry id="colour">
    <headword>colour</headword>
    <partOfSpeech tag="n"/>
    <label tag="europeanSpelling"/>
    <sense id="colour-1">
      <definition>
        <text>red, blue, yellow etc.</text>
      </definition>
      <example>
        <text>What is your favourite colour?</text>
      </example>
    </sense>
  </entry>
  <entry id="color">
    <headword>color</headword>
    <partOfSpeech tag="n"/>
    <label tag="americanSpelling"/>
  </entry>
  <relation type="variants">
    <member ref="colour"/>
    <member ref="color"/>
  </relation>
  <relationType type="variants">
    <description>words which differ only in spelling</description>
    <memberType type="entry" min="2" hint="navigate"/>
  </relationType>
</lexicographicResource>
```

Example A.81. JSON

```
{
  "uri": "http://example.com",
  "langCode": "en",
  "entries": [
    {
      "id": "colour",
      "headword": "colour",
      "partsOfSpeech": ["n"],
      "labels": ["europeanSpelling"],
      "senses": [
        {
          "id": "colour-1",
          "definitions": [{"text": "red, blue, yellow etc."}],
          "examples": [{"text": "What is your favourite colour?"}]
        }
      ]
    },
    {
      "id": "color",
      "headword": "color",
      "partsOfSpeech": ["n"],
      "labels": ["americanSpelling"]
    }
  ],
  "relations": [
    {
      "type": "variants",
      "members": [
        { "ref": "colour" },
        { "ref": "color" }
      ]
    }
  ],
  "relationTypes": [
    {
      "type": "variants",
      "description": "words which differ only in spelling",
      "memberTypes": [
        {
          "type": "entry",
          "min": 2,
          "hint": "navigate"
        }
      ]
    }
  ]
}
```

A.1.16 Modelling variants

We have two entries in our lexicographic resource, one for the headword "colour" and one for the headword "color". We want to create a relation to represent the fact that these are spelling variants. One of the entries is fully fleshed-out (has a sense with a definition, an example etc) while the other one is only skeletal: its only purpose is to serve as the origin of a navigable link to the other entry.

Example A.79. Pseudocode

```
- lexicographicResource (uri: http://example.com, langCode: en)
- entry (id: colour, headword: colour)
  - partOfSpeech (tag: n)
  - label (tag: europeanSpelling)
  - sense (id: colour-1)
    - definition (text: red, blue, yellow etc.)
    - example (text: What is your favourite colour?)
- entry (id: color, headword: color)
  - partOfSpeech (tag: n)
  - label (tag: americanSpelling)
- relation (type: variants)
  - member (ref: colour)
  - member (ref: color)
- relationType (type: variants, description: words which differ
  only in spelling)
- memberType (type: entry, min: 2, hint: navigate)
```

Example A.80. XML

```
<lexicographicResource uri="http://example.com" langCode="en"
xmlns="http://docs.oasis-open.org/lexidma/ns/dmlex-1.0">
  <entry id="colour">
    <headword>colour</headword>
    <partOfSpeech tag="n"/>
    <label tag="europeanSpelling"/>
    <sense id="colour-1">
      <definition>
        <text>red, blue, yellow etc.</text>
      </definition>
      <example>
        <text>What is your favourite colour?</text>
      </example>
    </sense>
  </entry>
  <entry id="color">
    <headword>color</headword>
    <partOfSpeech tag="n"/>
    <label tag="americanSpelling"/>
  </entry>
  <relation type="variants">
    <member ref="colour"/>
    <member ref="color"/>
  </relation>
  <relationType type="variants">
    <description>words which differ only in spelling</description>
    <memberType type="entry" min="2" hint="navigate"/>
  </relationType>
</lexicographicResource>
```

Example A.81. JSON

```
{
  "uri": "http://example.com",
  "langCode": "en",
  "entries": [
    {
      "id": "colour",
      "headword": "colour",
      "partsOfSpeech": ["n"],
      "labels": ["europeanSpelling"],
      "senses": [
        {
          "id": "colour-1",
          "definitions": [{"text": "red, blue, yellow etc."}],
          "examples": [{"text": "What is your favourite colour?"}]
        }
      ]
    }
  ]
}
```

Figure 4: Example 15

Published HTML (DocBook)	Markdown edition HTML (OASIS pipeline)
5.1.4.2 XML element: <entry> Implements the entry object type. Attributes • @homographNumber OPTIONAL Attributes if implementing the Linking Module • @id OPTIONAL. Entries that have an ID may use this ID as object reference instead of the IRI addressing mechanism described in Section 3.2.1, "DMLex fragment identification strings". Child elements • <headword> REQUIRED (exactly one). If implementing the Annotation Module <headword> can contain a mixture of plain text and placeholderMarker elements. • <partOfSpeech> OPTIONAL (zero or more) • <label> OPTIONAL (zero or more) • <pronunciation> OPTIONAL (zero or more) • <inflectedForm> OPTIONAL (zero or more) • <sense> OPTIONAL (zero or more) Child elements if implementing the Etymology Module • <etymology> OPTIONAL (zero or more) 5.1.4.3 XML element: <partOfSpeech> Implements the partOfSpeech object type. Attributes • @tag REQUIRED 5.1.4.4 XML element: <inflectedForm> Implements the inflectedForm object type. Attributes • @tag OPTIONAL Child elements • <text> REQUIRED (exactly one) • <label> OPTIONAL (zero or more) • <pronunciation> OPTIONAL (zero or more) 5.1.4.5 XML element: <sense> Implements the sense object type. Child elements • <indicator> OPTIONAL (zero or one) • <label> OPTIONAL (zero or more) • <definition> OPTIONAL (zero or more) • <example> OPTIONAL (zero or more) Attributes if implementing the Linking Module • @id OPTIONAL. Senses that have an ID may use this ID as object reference instead of the IRI addressing mechanism described in Section 3.2.1, "DMLex fragment identification strings". Child elements if implementing the Crosslingual Module • <headwordExplanation> OPTIONAL (zero or more) • <headwordTranslation> OPTIONAL (zero or more) 5.1.4.6 XML element: <definition> Implements the definition object type. Attributes • @definitionType OPTIONAL	5.1.4.2 XML element: <entry> Implements the entry object type. Attributes • @homographNumber OPTIONAL Attributes if implementing the Linking Module • @id OPTIONAL. Entries that have an ID may use this ID as object reference instead of the IRI addressing mechanism described in Section 3.2.1, "DMLex fragment identification strings". Child elements • <headword> REQUIRED (exactly one). If implementing the Annotation Module <headword> can contain a mixture of plain text and placeholderMarker elements. • <partOfSpeech> OPTIONAL (zero or more) • <label> OPTIONAL (zero or more) • <pronunciation> OPTIONAL (zero or more) • <inflectedForm> OPTIONAL (zero or more) • <sense> OPTIONAL (zero or more) Child elements if implementing the Etymology Module • <etymology> OPTIONAL (zero or more) 5.1.4.3 XML element: <partOfSpeech> Implements the partOfSpeech object type. Attributes • @tag REQUIRED 5.1.4.4 XML element: <inflectedForm> Implements the inflectedForm object type. Attributes • @tag OPTIONAL Child elements • <text> REQUIRED (exactly one) • <label> OPTIONAL (zero or more) • <pronunciation> OPTIONAL (zero or more) 5.1.4.5 XML element: <sense> Implements the sense object type. Child elements • <indicator> OPTIONAL (zero or one) • <label> OPTIONAL (zero or more) • <definition> OPTIONAL (zero or more) • <example> OPTIONAL (zero or more) Attributes if implementing the Linking Module • @id OPTIONAL. Senses that have an ID may use this ID as object reference instead of the IRI addressing mechanism described in Section 3.2.1, "DMLex fragment identification strings". Child elements if implementing the Crosslingual Module • <headwordExplanation> OPTIONAL (zero or more)

Figure 5: Section 5.1: XML serialization of entry

Appendix B References (Normative)

This appendix contains the normative and informative references that are used in this document. Normative references are specific (identified by date of publication and/or edition number or Version number) and Informative references are either specific or non-specific.

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B.1 Normative references (Normative)

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[RFC 3986] Tim Berners-Lee, Roy T. Fielding, Larry M Masinter Uniform Resource Identifier (URI): Generic Syntax, <https://datatracker.ietf.org/doc/rfc3986/> IETF (Internet Engineering Task Force) RFC 3986, January 2005.

[RFC 3987] Martin J. Dürst, Michel Suignard, Internationalized Resource Identifiers (IRIs), <https://datatracker.ietf.org/doc/rfc3987/> IETF (Internet Engineering Task Force) RFC 3987, January 2005.

[RFC 2119] S. Bradner, Key words for use in RFCs to Indicate Requirement Levels, <https://www.ietf.org/rfc/rfc2119.txt> IETF (Internet Engineering Task Force) RFC 2119, March 1997.

[RFC 8174] B. Leiba, Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words, <https://www.ietf.org/rfc/rfc8174.txt> IETF (Internet Engineering Task Force) RFC 8174, May 2017.

[BCP 47] M. Davis, Tags for Identifying Languages, <http://tools.ietf.org/html/bcp47> IETF (Internet Engineering Task Force).

[ITS] David Filip, Shaun McCance, Dave Lewis, Christian Lieske, Arle Lommel, Jirka Kosek, Felix Sasaki, Yves Savourel Internationalization Tag Set (ITS) Version 2.0, <http://www.w3.org/TR/its20/> W3C Recommendation 29 October 2013.

[JSON] The JavaScript Object Notation (JSON) Data Interchange Format, <https://tools.ietf.org/html/rfc8259> IETF RFC 8259 December 2017.

[RFC 3987] M. Duerst and M. Suignard, Internationalized Resource Identifiers (IRIs), <https://www.ietf.org/rfc/rfc3987.txt> IETF (Internet Engineering Task Force) RFC 3987, January 2005.

[RFC 7303] H. Thompson and C. Lilley, XML Media Types, <https://www.tools.ietf.org/html/rfc7303> IETF (Internet Engineering Task Force) RFC 7303, July 2014.

[Schematron] International Standards Organization, ISO/IEC 19757-3, Information Technology - Document Schema Definition Languages (DSDL) - Part 3: Rule-Based Validation — Schematron (Second Edition), http://standards.iso.org/ittf/PubliclyAvailableStandards/c055982_ISO_IEC_19757-3_2016.zip ISO, January 15, 2016.

[SQL] International Standards Organization, ISO/IEC 9075, Information technology – Database languages – SQL, <https://www.iso.org/standard/76583.html> ISO, 2023.

[UAX #9] M. Davis, A. Lanin, A. Glass, UNICODE BIDIRECTIONAL ALGORITHM, <http://www.unicode.org/reports/tr9/tr9-35.html> Unicode Bidirectional Algorithm, May 18, 2016.

[UAX #15] M. Davis, K. Whistler, UNICODE NORMALIZATION FORMS, <http://www.unicode.org/reports/tr15/tr15-44.html> Unicode Normalization Forms, February 24, 2016.

[Unicode] The Unicode Consortium, The Unicode Standard, <http://www.unicode.org/versions/Unicode9.0.0/> Mountain View, CA: The Unicode Consortium, June 21, 2016.

[XLIFF 2.1] David Filip, Tom Comerford, Soroush Saadatfar, Felix Sasaki, and Yves Savourel, eds. XLIFF Version 2.0, <http://docs.oasis-open.org/xliff/xliff-core/v2.1/os/xliff-core-v2.1-os.html> OASIS Standard 13 February 2018

[XML] W3C, Extensible Markup Language (XML) 1.0, <http://www.w3.org/TR/xml/> (Fifth Edition) W3C Recommendation 26 November 2008.

[XML namespace] W3C, Schema document for namespace <http://www.w3.org/XML/1998/namespace> <http://www.w3.org/2001/xml.xsd> <http://www.w3.org/2009/01/xml.xsd>.

[XML Catalogs] Norman Walsh, XML Catalogs, <https://www.oasis-open.org/committees/download.php/14809/xml-catalogs.html> OASIS Standard V1.1, 07 October 2005.

[XML Schema] W3C, XML Schema, refers to the two part standard comprising [XML Schema Structures] and [XML Schema Datatypes] (Second Editions) W3C Recommendations 28 October 2004.

[XML Schema Datatypes] W3C, XML Schema Part 2: Datatypes, <http://www.w3.org/TR/xmlschema-2/> (Second Edition) W3C Recommendation 28 October 2004.

[XML Schema Structures] W3C, XML Schema Part 1: Structures, <https://www.w3.org/TR/xmlschema-1/> (Second Edition) W3C Recommendation 28 October 2004.

[DCMI Metadata Terms] DCMI Usage Board, DCMI Metadata Terms, <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/> Date issued: 2020-01-20.

B.2 Informative references (Informative)

[NVH] Name-Value Hierarchy, <https://www.namevaluehierarchy.org/>.

[LDML] Unicode Locale Data Markup Language <http://unicode.org/reports/tr35/>

[UAX #29] M. Davis, UNICODE TEXT SEGMENTATION, <http://www.unicode.org/reports/tr29/> Unicode text Segmentation.

Appendix C Machine Readable Validation Artifacts (Normative)

- [XML schema](#)
- [XML schema excluding the cross-lingual module](#)
- [JSON schema](#)

Appendix B References (Normative)

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[RFC 3987] Martin J. Dürst, Michel Suignard, Internationalized Resource Identifiers (IRIs), <https://datatracker.ietf.org/doc/rfc3987/> IETF (Internet Engineering Task Force) RFC 3987, January 2005.

[RFC 2119] S. Bradner, Key words for use in RFCs to Indicate Requirement Levels, <https://www.ietf.org/rfc/rfc2119.txt> IETF (Internet Engineering Task Force) RFC 2119, March 1997.

[RFC 8174] B. Leiba, Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words, <https://www.ietf.org/rfc/rfc8174.txt> IETF (Internet Engineering Task Force) RFC 8174, May 2017.

[BCP 47] M. Davis, Tags for Identifying Languages, <http://tools.ietf.org/html/bcp47> IETF (Internet Engineering Task Force).

[ITS] David Filip, Shaun McCance, Dave Lewis, Christian Lieske, Arle Lommel, Jirka Kosek, Felix Sasaki, Yves Savourel Internationalization Tag Set (ITS) Version 2.0, <http://www.w3.org/TR/its20/> W3C Recommendation 29 October 2013.

[JSON] The JavaScript Object Notation (JSON) Data Interchange Format, <https://tools.ietf.org/html/rfc8259> IETF RFC 8259 December 2017.

[RFC 3987] M. Duerst and M. Suignard, Internationalized Resource Identifiers (IRIs), <https://www.ietf.org/rfc/rfc3987.txt> IETF (Internet Engineering Task Force) RFC 3987, January 2005.

[RFC 7303] H. Thompson and C. Lilley, XML Media Types, <https://www.tools.ietf.org/html/rfc7303> IETF (Internet Engineering Task Force) RFC 7303, July 2014.

[Schematron] International Standards Organization, ISO/IEC 19757-3, Information Technology - Document Schema Definition Languages (DSDL) - Part 3: Rule-Based Validation — Schematron (Second Edition), http://standards.iso.org/ittf/PubliclyAvailableStandards/c055982_ISO_IEC_19757-3_2016.zip ISO, January 15, 2016.

[SQL] International Standards Organization, ISO/IEC 9075, Information technology – Database languages – SQL, <https://www.iso.org/standard/76583.html> ISO, 2023.

[UAX #9] M. Davis, A. Lanin, A. Glass, UNICODE BIDIRECTIONAL ALGORITHM, <http://www.unicode.org/reports/tr9/tr9-35.html> Unicode Bidirectional Algorithm, May 18, 2016.

[UAX #15] M. Davis, K. Whistler, UNICODE NORMALIZATION FORMS, <http://www.unicode.org/reports/tr15/tr15-44.html> Unicode Normalization Forms, February 24, 2016.

[Unicode] The Unicode Consortium, The Unicode Standard, <http://www.unicode.org/versions/Unicode9.0.0/> Mountain View, CA: The Unicode Consortium, June 21, 2016.

[XLIFF 2.1] David Filip, Tom Comerford, Soroush Saadatfar, Felix Sasaki, and Yves Savourel, eds. XLIFF Version 2.0, <http://docs.oasis-open.org/xliff/xliff-core/v2.1/os/xliff-core-v2.1-os.html> OASIS Standard 13 February 2018

[XML] W3C, Extensible Markup Language (XML) 1.0, <http://www.w3.org/TR/xml/> (Fifth Edition) W3C Recommendation 26 November 2008.

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[XML Catalogs] Norman Walsh, XML Catalogs, <https://www.oasis-open.org/committees/download.php/14809/xml-catalogs.html> OASIS Standard V1.1, 07 October 2005.

[XML Schema] W3C, XML Schema, refers to the two part standard comprising [XML Schema Structures] and [XML Schema Datatypes] (Second Editions) W3C Recommendations 28 October 2004.

[XML Schema Datatypes] W3C, XML Schema Part 2: Datatypes, <http://www.w3.org/TR/xmlschema-2/> (Second Edition) W3C Recommendation 28 October 2004.

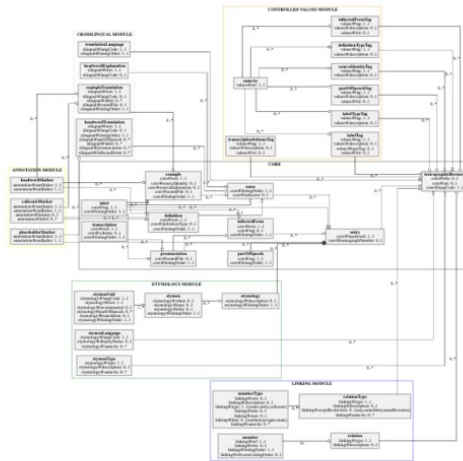
[XML Schema Structures] W3C, XML Schema Part 1: Structures, <https://www.w3.org/TR/xmlschema-1/> (Second Edition) W3C Recommendation 28 October 2004.

[DCMI Metadata Terms] DCMI Usage Board, DCMI Metadata Terms, <https://www.dublincore.org/specifications/dublin-core/dcmi-terms/> Date issued: 2020-01-20.

B.2 Informative references (Informative)

Figure 6: Appendix B References

Appendix D DMLex UML diagram (Normative)

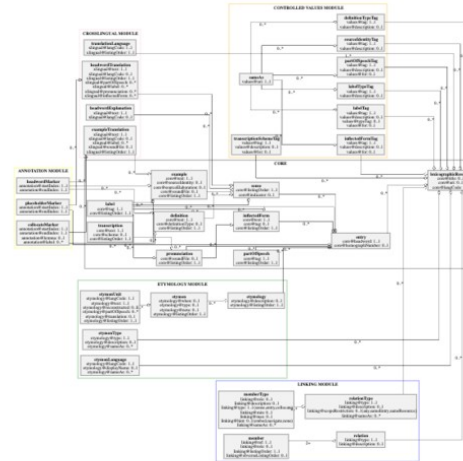


Appendix E DMLex plain text description (Informative)

This description uses the [NVH schema notation style](#) and the convention that all object types and their properties are prefixed by either `core@` or `MODULENAME@` strings. Their arity is denoted by a * (zero or more), + (one or more), ? (zero or one) or N+ (N or more). If no arity is given, it indicates the arity of 1, i.e. being required and occurring exactly once.

```
core@lexicographicResource:
  core@title: ?
  core@uri: ?
  core@langCode:
  core@entry: *
    xlingua@translationLanguage: +
    values@definitionTypeTag: *
    values@inflectedFormTag: *
    values@labelTag: *
    values@labelTypeTag: *
    values@partOfSpeechTag: *
    values@sourceIdentityTag: *
    values@transcriptionSchemeTag: *
    linking@relation: *
    linking@relationType: *
    etymology@etymonLanguage: *
    etymology@etymonType: *
  core@entry:
    core@headword:
      core@homographNumber: ?
      core@partOfSpeech: *
      core@label: *
      core@pronunciation: *
      core@inflectedForm: *
      core@sense: *
      annotation@placeholderMarker: *
      etymology@etymology: *
    core@partOfSpeech:
      core@tag:
      core@listingOrder:
    core@inflectedForm:
      core@text:
      core@tag: ?
      core@label: *
      core@pronunciation: *
      core@listingOrder:
    core@sense:
      core@listingOrder:
      core@indicator: ?
      core@label: *
      core@definition: *
      core@example: *
      xlingua@headwordExplanation: *
      xlingua@headwordTranslation: *
    core@definition:
      core@text:
      core@definitionTypeTag: ?
```

Appendix D DMLex UML diagram (Normative)



Appendix E DMLex plain text description (Informative)

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```
core@lexicographicResource:
  core@title: ?
  core@uri: ?
  core@langCode:
  core@entry: *
    xlingua@translationLanguage: +
    values@definitionTypeTag: *
    values@inflectedFormTag: *
    values@labelTag: *
    values@labelTypeTag: *
    values@partOfSpeechTag: *
    values@sourceIdentityTag: *
    values@transcriptionSchemeTag: *
    linking@relation: *
    linking@relationType: *
    etymology@etymonLanguage: *
    etymology@etymonType: *
  core@entry:
    core@headword:
      core@homographNumber: ?
      core@partOfSpeech: *
      core@label: *
      core@pronunciation: *
      core@inflectedForm: *
      core@sense: *
      annotation@placeholderMarker: *
      etymology@etymology: *
    core@partOfSpeech:
      core@tag:
      core@listingOrder:
    core@inflectedForm:
      core@text:
      core@tag: ?
```

Figure 7: Appendix D UML diagram



Data Model for Lexicography (DMLex) Version 1.0

OASIS Standard

29 April 2025

Specification URIs

This version:

<https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/dmlex-v1.0-os.html>
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Additional artifacts:

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- RDF: <https://docs.oasis-open.org/lexidma/dmlex/v1.0/os/schemas/RDF/>

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dmlex-v1.0-os

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Figure 9: PDF table of contents

Example 5. NVH

```
lexicographicResource: http://example.com/dictionary
  title: An example dictionary
  langCode: en
  entry: ...
  entry: ...
```

3.4 entry

Represents a dictionary entry. An entry contains information about one headword.

Property of

- lexicographicResource

Properties

- headword REQUIRED (exactly one) and UNIQUE (in combination with other unique properties if present). Normalised string. The entry's headword.
- partOfSpeech OPTIONAL (zero or more) and UNIQUE (in combination with other unique properties if present).
- homographNumber OPTIONAL (zero or one) and UNIQUE (in combination with other unique properties if present). Number. The entry's homograph number, as a guide to distinguish entries with the same headword.
- label OPTIONAL (zero or more).
- pronunciation OPTIONAL (zero or more).
- inflectedForm OPTIONAL (zero or more).
- sense OPTIONAL (zero or more).

Note

DMLex Core does not have a concept of "subentry". To model subentries (i.e. entries inside entries) in a lexicographic resource, object types from the Linking Module should be used.

Note

The headword can be a single word, a multi-word expression, or any expression in the source language which is being described by the entry.

Note

DMLex allows only one headword per entry. Things such as variant headwords do not exist in DMLex. However, the DMLex Linking Module does make it possible to represent the existence of variants by treating them as separate headwords of separate entries, and linking the entries using a type of link which will cause the entries to be placed together when shown to human users. See Section A.1.16, "Modelling variants" for an example using the English words "colour" and "color".

Example 1. XML

```
<lexicographicResource uri="..." langCode="...">
  <title>...</title>
  <entry.../>
</lexicographicResource>
```

Example 2. JSON

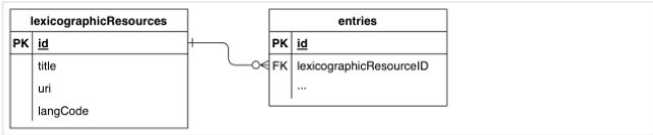
```
{
  "title": "...",
  "langCode": "...",
  "entries": [...]
}
```

Example 3. RDF

```
@prefix dmlex:
<https://docs.oasis-open.org/lexidma/dmlex/v1.0/schemas/RDF/dmlex.ttl#> .

<#id> a dmlex:LexicographicResource ;
dmlex:title "...";
dmlex:uri "...";
dmlex:langCode "...";
dmlex:entry <entry1> , ... .
```

Example 4. Relational database



Example 5. NVH

```
lexicographicResource: http://example.com/dictionary
  title: An example dictionary
  langCode: en
  entry: ...
  entry: ...
```

3.4 entry

Represents a dictionary entry. An entry contains information about one headword.

Property of

- lexicographicResource

Properties

Figure 10: PDF section 3.4 entry

Note

DMLex allows entries to contain zero senses. An entry with zero senses is an entry which declares the existence of its headword and, optionally, gives some information about the headword's morphosyntactical properties (part of speech, inflected forms etc.) but does not say anything about its semantics, about its meaning. Such entries can exist as entries which are work in progress, or as entries whose only purpose is to serve as members in entry-to-entry relations (defined in the [DMLex Linking Module](#)) such as spelling variants (example: [Section A.1.16, "Modelling variants"](#)).

Note

Entries in DMLex do not have an explicit listing order. An application can imply a listing order from a combination of the headword and the homograph number, such that the headword is the primary sorting key and the homograph number (for entries that have one) is the secondary sorting key.

Note

Ideally, each entry should have exactly one part-of-speech label. However, DMLex allows more than one `partOfSpeech` in `entry` in order to allow for exceptional cases when the lexicographer has decided to treat multiple part-of-speech readings of a headword in a single entry. Example: English words which denote nationalities ("Czech", "German") and which can function both as nouns and as adjectives.

Example 6. XML

```
<entry homographNumber="...">
  <headword>...</headword>
  <partOfSpeech.../>
  <label.../>
  <pronunciation.../>
  <inflectedForm.../>
  <sense.../>
</entry>
```

Example 7. JSON

```
{
  "headword": "...",
  "homographNumber": "...",
  "partsOfSpeech": [...],
  "labels": [...],
  "pronunciations": [...],
  "inflectedForms": [...],
  "senses": [...]
}
```

Ideally, each entry should have exactly one part-of-speech label. However, DMLex allows more than one `partOfSpeech` in `entry` in order to allow for exceptional cases when the lexicographer has decided to treat multiple part-of-speech readings of a headword in a single entry. Example: English words which denote nationalities ("Czech", "German") and which can function both as nouns and as adjectives.

Example 6. XML

```
<entry homographNumber="...">
  <headword>...</headword>
  <partOfSpeech.../>
  <label.../>
  <pronunciation.../>
  <inflectedForm.../>
  <sense.../>
</entry>
```

Example 7. JSON

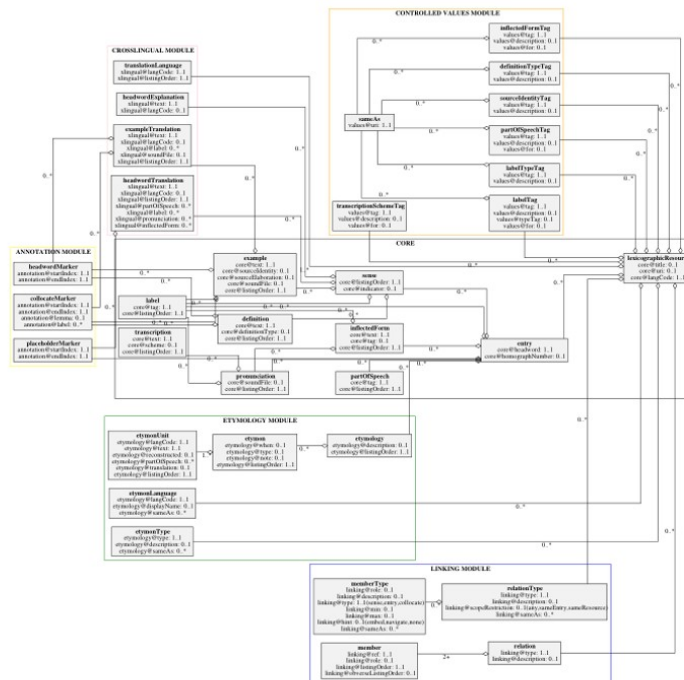
```
{
  "headword": "...",
  "homographNumber": "...",
  "partsOfSpeech": [...],
  "labels": [...],
  "pronunciations": [...],
  "inflectedForms": [...],
  "senses": [...]
}
```

Example 8. RDF

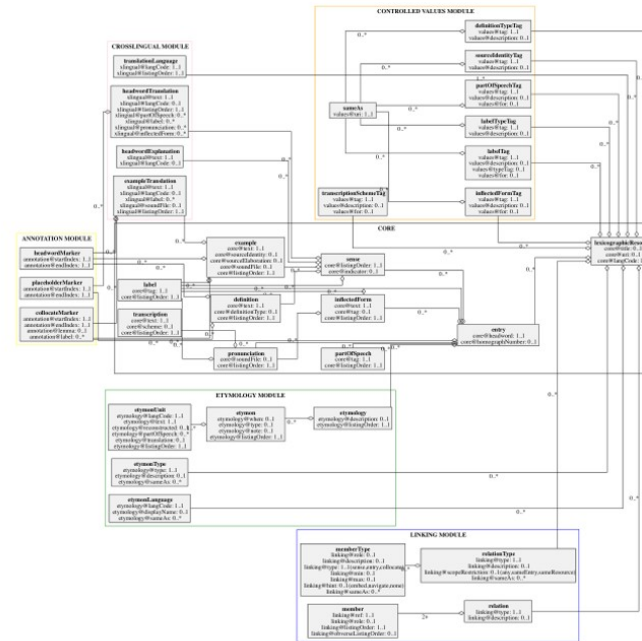
```
<id> a dmlex:Entry ;
dmlex:headword "...";
dmlex:homographNumber ... ;
dmlex:partOfSpeech ... ;
dmlex:label ... ;
dmlex:pronunciation ... ;
dmlex:inflectedForm ... ;
dmlex:sense ... .
```

Figure 11: PDF Example 6

Appendix D DMLex UML diagram (Normative)



Appendix D DMLex UML diagram (Normative)



Appendix E DMLex plain text description (Informative)

This description uses the [NVH schema notation style](#) and the convention that all object types and their properties are prefixed by either `core` or `MODULENAME` strings. Their arity is denoted by a * (zero or more), + (one or more), ? (zero or one) or N+ (N or more). If no arity is given, it indicates the arity of 1, i.e. being required and occurring exactly once.

```
core@lexicographicResource:
  core@title: ?
  core@uri: ?
  core@langCode:
  core@entry: *
  xlingual@translationLanguage: +
  values@definitionTypeTag: *
  values@inflectedFormTag: *
  values@labelTag: *
  values@labelTypeTag: *
  values@partOfSpeechTag: *
```

Figure 12: PDF Appendix D UML diagram